

ActInf Livestream #023.2 ~ "Embodied skillful performance: where the action is"

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greetings everyone welcome to actin flab live stream number 23.2 today is june 15th 2021 and we're going to be having our dot 2 discussion on embodied skillful performance where the action is and we'll see who else joins us live as we go through the slides in the paper and also of course if you're watching live we'd welcome any questions or comments welcome to the active inference lab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links that are on this slide this is recorded greetings dave you this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams we're again on june 15th having our second group discussion on embodied skillful performance where the action is so this is sort of our jumping off point and let's see if we can make it a skillful one today the goals are really just to be discussing and learning to be hearing the questions and points that people found really interesting when they were reading the paper as well as see where our discussion can spontaneously go as of course participants join and leave and hopefully we get some fun questions from the chat and we're just going to walk through the paper and the slides go where we need to go to hopefully connect some dots so with that being said let's just go to the introductions and warm-ups and we can each just give an introductory thought on the paper if we would like or something we liked or remembered something we're looking to resolve reduce our uncertainty on by the end of the discussion so i'm daniel i'm a postdoctoral researcher in california and all passed to dean hi i'm dean i'm in calgary and i'm becoming a more embedded personality in this active inference lab world and i'll pass it over to stephen hello i'm stephen i'm based in toronto i'm doing a practice based phd through the uk i'm interested in um participatory theater community development and ways to understand how people interact with each other uh in groups and individually so this paper is quite relevant as it opens up some new avenues in that and i will pass it over to is it is dave here dave hello

dave yeah dave if you can speak otherwise no worries maybe you're just listening in so maybe just a quick question stephen where do we see the connection between skillful performance which is often thought of like motor and bodily how do we connect that to social practices which are amongst bodies and use verbal behavior or symbolic behavior yeah i mean this is this is kind of a question i think that's arising more and more as people are trying to question psychology or community psychology which works on the individual or sociology which works on this sort of mass of people but when you want to work with intersubjectivity and sort of small groups and understanding the dynamics that are going on there you could kind of ask phenomenological questions about what it's like to be in that state which is kind of a bit more direct than say someone's psychological perspective on it but it's still not quite working so then you say well let's get into action here let's get into what's actually happening in and amongst people um and often if you then jump into action that's another abstracted area but if you're in the area of practice um you're kind of in a in between you know it's something that people are doing something that people are doing between each other and but you're still able to talk about how it's happening how it feels um so i think that um that this this question of practice then you know takes you into the sort of areas that get used a lot in arts arts practice theater practice all these types of areas so um i think it's quite a a nice nexus that's normally a bit impenetrable cool it makes me wonder do groups have skill or is skill something that only bodies have so you see some football team or basketball team do some very coordinated play or two dancers that are improvising is that able to be reduced entirely to the skilled embodied performance of individual bodies or is there a way in which we could talk about a group embodying skillful performance especially because so many performances i mean except for what stand-up comedy a few other things their group and then even those individual performances are in feedback sometimes with a group it's just different roles um that's a good point actually i mean i think that's really the exciting field of practice that's going to be growing over the next while and i i sense that it's almost impossible to get that from a snapshot you have to have something that's been progressing over time to get or let's just say i don't know whether that's true but i i sense that there's more use in understanding how things change over time and getting the dynamics than there is trying to analyze a snapshot uh dave if you have any comments you can unmute and i'll see it or dean any sort of just opening thoughts how does this dot 2 meet you today well i'm going to come from it from a little different angle i so i think that word between is a huge is it it's the thing that that befuddles us and yet it's the thing that we probably want to know the most about and um in this idea of embodied skill for

skillful performance i was going to actually ask the author some questions around how she relates it back to sort of the parts of the brain that um resolve these problems so is it a spatial relationship thing like what the hippocampus tends to provide us some insight on or is it a caudate nucleus thing which says that we sequence everything out and we turn left we turn right we release the ball and i was just curious about which because those two those two things work but when one is on the other one is off there's kind of a binary relationship between those two parts of the brain and so i wanted to kind of figure out how that that toggling back and forth is part of this skill that we are witnessing nice point there's um the more cognitively driven versus almost the more bodily driven and then that handoff is where we see skillful performance and that's where we see choking it's where we see all these dynamics between cognitive processes and action processes okay we see another dave dave if you can if you want to give any thoughts just feel free um yeah oh just listening and thanks nice right on that's the real that's the real og way to listen in in the jitsi rather than in the live stream um yeah so we'll see who else joins in today but that's a great question about the brain pieces and then i always think about other species like if insects can have embodied action and that seems pretty fair to say can they have skillful action and how would we know and then if they do the layout of their brain and body is so different that it wouldn't map on you know they don't have a hippocampus they don't have a prefrontal cortex they have other brain regions so we don't need to get too locked into one implementation of these processes based upon human neuroanatomy when we know that there's other neural anatomies that can deploy similar processes so it helps us keep an open mind the multiple realizations of active inference or of embodied skillful performance because of just the diversity of evolutionary systems let's take an overview at the roadmap this paper has a form that we've seen several times before which is in the first half taking some non-active inference body of research so in this case it's skillful performance and optimal motor control theory and investigating with a critical lens this previous body of research to understand what is it really saying or what are the assumptions so in this case there's this instructionist assumption implicitly or explicitly that's drawn out through the exploration of previous approaches to skillful performance and specifically the motor representations and motor commands that are implicit in optimal motor control theory so that's part one part two goes from optimal control to predictive coding and active inference and last time we talked a little bit about similarities and differences as you go from optimal motor control to predictive processing to active inference what is generalized as you move from a to b to c what is different what kinds of issues might you stick or introduce and then in

the last sections of the paper motor control is positively reframed as this interactive sensory motor engagement with the world which puts us very squarely in the domain of active inference rather than in the instructionist domain of optimal motor control theory greetings blue so that's kind of a general pattern that we've seen before where the first third of the paper is a background on previous approaches in a critical lens the second approach shows some sort of a transition from theory a to b to c and then by the third part there's a positive positivist argument as to why active inference is more in line with how things are or might be more useful uh steven go ahead so yeah further to our conversation earlier um i think this actually this this showing the instructionist and then the approach is really useful to sort of have a plausible way for affect and social dynamics to interplay with action because when it's under an instructionist model it is kind of hard to imagine how that kind of mechanical almost message passing type of approach can integrate into our sort of social dynamics whereas um the interaction in this approach does give that so i think that's quite good it can frighten people off because it brings in this other philosophical piece which i think a lot of people would like running away from at first but actually it does offer a more humorous humanizing or biological way to engage cool let's just go there blue of course raise your hand anytime you want to add something but this instructionism versus interactionism i think is a big piece that we'll walk away from this uh 23 width it's something that we can take on in the case of embodied skillful performance but i think we're going to be able to apply it to education we're going to be able to apply it to team collaboration and so we can definitely return to this distinction between instructionism and interactionism so steven just from you know your experience with drama or other kinds of work do you think people are how do people implicitly use instructionism versus interactionism or what kinds of scenarios uh support or constrain instructionism versus allow for interactions to emerge well i think what tends to happen is in the models that people tend to use they're kind of either in one camp or another it's like so mostly it's almost implicit that there's some sort of interactive stroke flow it probably goes even further it's sort of some sort of flow in the sort of performance arts world they talk about flow and ease and that type of thing and what's also interesting is i know they talk about this in the michael chekhov acting approach is there's always this battle between ease and form so a dancer gets more form they get more technique they get more proficient their technique but they might lose their ease that you have as a beginner so you're forever trying to add more technique without becoming this kind of rigid autumn automaton you know so i think there's there's that kind of side to it and and that that whole area they just go off and

it's it's kind of it just takes care of itself but it becomes a little bit um of a silo in a way and then you've got people working in more straight activity theory veikovsky kind of much more like what activity happened when um and that they they they are also in a bit of a silo so i think that you tend to find that and different different areas they they are a little bit siloed in their approaches um to sort of get around some of these um challenges um so i think that the way of trying to break beyond what you can analyze which often you have to look at instructionally because you kind of have to break down things into steps but that doesn't mean you need to keep it in a stepwise form in terms of how we understand what people really do so there's this kind of i think there's this kind of gap that this will help to make plausible interesting there how we capture the utility and the apparent advantages of instructionism some of them are written here like the ability to separate different instructions that seems pretty important if you say well how am i going to cook this multi-stage dish and someone just says you know here's all the ingredients and here's all the behaviors but i'm not going to tell you what order to do what in if we can't separate and sequence events then it's going to be hard to carry out a performance correctly you can't just switch the order of musical notes and the sheet music you can't just do things in a different order so how do we get some of the features that we're looking for like separability and modularity and take it into a fully interactionist framework one distinction there that the authors raise is like strong versus weak instructionism and they'll um end up continuing that both of these forms have similar underpinnings but there's sort of uh a continuum between thinking that the instructionism is completely its own neural representationism it's kind of like a mott and bailey type argument this is the strong instructionism thinking that it's totally just the neural representations that are entirely specifying on their own the movements that have to be performed a more dynamical coupling version of instructionisms so that'd be like you know when you're tapping the gas it's like an instruction to the engine but it's not the entire um go faster cue given the background of ecologically normal processes that enable them to play this role so that's kind of interesting um blue any sort of overarching thoughts or anywhere you think it'd be fun to jump in no i'm just trying to figure out where i'm at still indeed um well there'll be a few other places like a few other notes that we wanted to discuss dean mentioned um one thing maybe we can even have a dot three and find out what a dot three would uh mean or dot two point dot two one or something like that dot twenty one um so one piece was about the the neural elements because in the figures we do see brains however um these are quite schematic it's not saying that these are the brain regions where these types of things happen which can be a little uh ambiguous

because it's not actually an anatomical representation yet it's superimposed on top of a brain
um okay so one piece we wrote down was about the neural and i think there was a few um
yeah oh yeah stephen go for it yeah just tying this as you mentioned about the streak the weak
and strong um it's kind of interesting because that's kind of trying to give this scientific feel
but it doesn't feel very satisfactory to try and have to say it in terms of weak and strong
because you're kind of in this well what does that really mean and how does it shift between
the two i mean i can understand what has been said but um it's it gets into this kind of fuzzy
version of instructionism so i i think that kind of sets up some of these questions that as soon
as you move away from a strong instructionism um it becomes harder to hold it together using
the word instructionism so i think that's it's quite a good illustration of that because it's sort of
unraveled you can see how the second paragraph's a lot longer than the first one basically yep
great point and it's also a common approach to basically rigorous philosophical argument
instead of making an argument against a total category you introduce a distinction in the
category and then you might find that either one of those categories has some feature that the
other one doesn't you know instead of being against all wars well there's the good ones and
there's the bad ones and then here's my distinction for just the bad ones or it could be that by
making that distinction you then can say actually this holds true for both of these categories or
subcategories and so that's how i've made my claim against all x is by first breaking it down
and then they do again find that there of course are differences between these two forms of
instructionism but that they end up taking on some of the same assumptions um cool there's
probably a few other pieces we can explore this is the section in the paper that i'm sharing on
the screen that says the formulation of sensory motor control in terms of heavily hinges on two
different but highly interconnected assumptions so let's see what the assumptions are because
it's probable that active inference is going to be contrasting or doing something different so
one is the central specification of descending motor commands and their efferent copies in the
form of detailed low-level instructions for control of the motor plant specified in terms of an
intrinsic frame of reference i.e extension and contraction of muscle fibers so that's that
instructionism and then two a separation of forward and inverse models operating on
complementary aspects of action planning and executionism so the first piece is that
instructionism that what's being sent descending or towards the body are instructions and the
second piece is a separation of the forward and the inverse models operating on
complementary aspects of action planning and execution so the forward model is like sort of a
planning model and then the execution is like okay given what i planned how am i going to

execute that and those are two different streams in optimal motor control and so we've been talking about how interactionism we can explore how active inference deploys interactionism but in interactionism we have a contrast with the instructionist point one and then through a unified generative model in active and the reduction of uncertainty in a unified generative model we're going to move beyond this separation of these two forward and inverse models stephen and then anyone else or any questions in the live chat yeah this is quite helpful it gives that realization that optimal motor control theory it could it can be very precise you could imagine someone making a robot because you can measure everything you've got access to all the data and you could make something within like a thousandth of a millimeter precision which humans or animals can't do but it it does have a bit of a limitation in terms of um being able to integrate and i think that i don't know if you've seen any of the papers around the idea of fields being around the robot's joints i have like a field around the hand a field around the and just move into the field rather than trying to work out what the actual must like the actual joints should do and they they out competed other robots within like two or three months so the same idea of working with some sort of more fuzzy um dynamical field actually and just working out what works most efficiently or best even though you're probably adding a lot more fuzziness into the actual understanding of the measurements actually has a lot of them has a lot of benefits in terms of learning how to do something quickly okay i think there'll be a few more cool points to return to on this notion of field but i think it's also helpful here to look at how they distinguish the forward and the inverse models so here a forward model takes a system from an intrinsic to an extrinsic frame predicting the effects of different movements using musculoskeletal plans specified by neural activity so again that's that instructionism that's because we're in the optimal motor control so forward model is basically plans that can be cognitive into implementations that's the translation into motor commands and their consequences on the other hand an model builds motor commands by inverting the causal change so first that inverse model leverages a value function of states this is where we get the inoptimal motor the non-negotiable introduction of value as a central heuristic states have to be assigned value because value is what's used to determine which plans to select and that leveraged on the value function is used to form a mapping from desired target states which again is built into active inference in a sense through the preference over certain types of states but the desired target state has to be specified in a slightly different way as an actual preferred state rather than over a preferences over states in this optimal motor control framework and that's in the coordinate system based upon the external consequences of so i want my my arm

to be at a 90 degree angle and then that is translated or it's mapped to a set of intrinsic coordinates in the space of muscle fiber activations like my arm is at a greater than 90 degree angle so that needs to map to activation of bicep firing patterns so that the arm can contract so that the angle is smaller and that's why frameworks based upon optimal motor control are sometimes characterized in terms of force control because all the motor commands specify is muscle forces and joint torques so that's kind of the macro to micro mapping is we have preferred states and the question is there's so many degrees of freedom with the joints so how do we go from potentially a broad set of preferred states or acceptable states into a specific pattern of muscular activations so how do you go from make your hand just rest where it's comfortable which is a relatively broad sort of plateau of accessible states into a position that each joint is going to be at equilibrium with and the two of these assumptions that we've been exploring here so again one the instructionism and two the separation of forward and inverse model that architecture induces the assumption or rests on the assumption that value valuable states is what causes action and that's what that first in 2011 paper introduced is sort of flipped that on its head which is that well um in models from omct sequences of action are selected according to the value function of their states so value is something that's cognitive and then induces action but that gets sort of flipped in the 2011 perspective because value is a function of the states not a cause of them so as written here value is an attribute of states that are caused by movement it's a consequence not a cause and when we get loops you know which one's a cause of which something that's locally a causes b but you pull back a level and maybe the relationship can also be seen as sort of the other direction or maybe with feedback systems or embodied systems we move beyond just any single linear framing of what causes what but regardless of the direction or whether value is an attribute of states or it's a consequence or it's a cause the point is there still is this central value parameter in optimal motor control theory and that value parameter which is familiar to reinforcement learning like learning preferable states based upon their predicted value that's something that we're going to be able to build on in active inference by pursuing not just value but also allowing for reduction of uncertainty in a generative model which can include pragmatic or value-driven elements but also elements that include curiosity like drives steven and then anyone else and this ties in with kasper hess work with affect you could imagine that the affective states are kind of generated on the fly at different scales so as you start to move maybe slowly as you you see that these value states are kind of almost revealed and not necessarily available and then maybe there's a state for once the whole action is complete and that gives you you know enter that value state which maybe

only appears as that um degree of surprisal is realized and it's always going to be slightly different each time you do something so this this is quite cool i've moved here to that kind of vector field representation of the difference between value only learning and performance here's our robot who's only interested in kind of taking the ruler putting it on the hill and then going in the direction of the ruler going uphill that's the sort of scalar approach to pursuing peaks in contrast we have that peak climbing or hill climbing element in active inference as that irrotational or longitudinal current but we also get this solenoidal element so i'm wondering talking to people about performance for example or training in some way how would it be different to promote this sort of value and equivalent flow model rather than a comparative reward driven model like if a learner is constantly assessing whether it's a multiple choice test or whether it's baseball pitching or something like that if a learner is always assessing the value of states and everything is framed as value maximization and you're going to want to be doing accurate states you want you know the pitch you want to be getting your pitches more accurate because more accurate is more rewarding i'm just wondering how it changes training or the experience of practice to know that there's times when one is pursuing value or that in the long range pursuit of value can be achieved through this rotational or exploratory element blue and then anyone else so just talking about pitching specifically like yeah you can you know hit the bullseye right if you're gonna pitch you know pitch a ball you can hit the bullseye you can hit the target get it right over home plate but like you never explore like what about like spinning the ball what effect will like different spins that you put on the ball have on you know the way that the batter hits it so like the accuracy versus like the finesse of it and i think about that in terms of dance also like if you're not if you don't explore like you know what it feels like like if you're only trying to accurately make a movement you're never exploring like the finesse element of it so there's no like room for perfection or like ultimate mastery i think and where does skill become mastery it's sort of like at least in chess it's the the masters who appear to challenge some of the basic rules like they don't castle early because there's some deeper plan or a jazz musician who moves beyond the framework of regular timing to introduce some other element so that play element becomes especially important it's almost like if you never learn how to do it correctly or normatively you can't throw the baseball so you got to get it over the plate but then the mastery level after performance has been demonstrated in a sense involves inventing a new way to pitch like there's you know a fraction of baseball pitchers who throw sidearm or throw knuckleball all these different sort of unconventional approaches that aren't the value function that they

learned initially or maybe it was their biomechanics and that was the value function that worked for them and some other value function just wouldn't have worked with their body and brain steven and then anyone else yeah thinking from the perspective of coaching because um so my supervisor works a lot with systemic coaching and performance coaching is this this becomes a more unifying approach it might be for specific things optimal control theory is quite useful for just unpacking something but if you want to look at performance coaching more broadly or developmental coaching more broadly and how they relate this could be useful and i'm just thinking about christian ericsson the you know the soccer player i know if you heard he collapsed during the euros and uh his heart stopped and all the team were around him trying to and then they had to play the game in about an hour's time now their ability to perform um was going to be very hard it's not a performance coaching question at that point there's these other dynamics going on and i think that this this ability for active inference to at least unify how regimes can change and how it could be the environment the system the the emotions all these things are quite um quite useful thanks dean yeah yeah i appreciate what you're saying there steve and i i am aside from the from the step-by-step part that i think the caudate nucleus tends to to perform when we're when we're trying to climb that little mountain there robotically it's another factor i think that's kind of interesting and that's um if it's embodied and it's skilled there's probably three kinds of math at play there's materialization the how sort of the engineering of it there's the what's in what's out what's supliferous and what what needs to be an element within the performance and then i think what this paper does in the interactions piece pieces it brings the the probably into the conversation and i think you need all three kinds of calculations in order to be able to get a real grip on on what's going on maybe not an optimal grip at least a fuzzy grip on what to do next i think if you only have two of those elements you've got the perfect conditions for an instrumentalist approach when you add the probably whether it's erickson um what did he text or email all of his his teammates and said go finish the game we gotta we gotta finish this game and i think steven's right you can you can give an instruction to your teammates and say i'm fine but you just had a cardiac arrest and we probably aren't going to be able to wipe that from our memory in that moment and so it's a little bit more complicated when we add probably but i think we need probably in order for it to be interactionist so that team like in the middle of the game the personnel of the team changed in the individual story of motor control it's like but that's not between you and the soccer ball so why should it change performance but when we have the deep generative model of the individual that includes affect that's one way to get to the loss of

a teammate influencing performance and the second way would be to actually genuinely allow teams to be skillful performers in the sense that there's a distributed cognitive process on the team just like there is in the body and so they actually changed and whoever steps that person's position on the field they're gonna have a different approach so it'd be like switching out you know one arm for another arm or one forager nest mate for another nest estimate it's going to change the emergent properties of the group in a way that's an explanation at the group level rather than requiring to come back to just the optimal motor control value driven instruction guided of a body blue yeah just the probably is like what gives active inference its power right like probably if i don't eat breakfast i'm going to collapse in the middle of a of a field you know or playing a game or whatever um so so when you have that that probably it it allows you to guide all of your actions toward a given intention i think thanks blue steven yeah and i also think when probably because probably often it's still in that venue my world you know probably i will get this game or that game when the word uncertainty is put in uncertainty is a much scarier word than probability right because it's like it has that risk component um so it's really interesting that you say that that that um we tend to say probably everything will be okay but active influence is like you don't know it's all it's all unknown to some extent here's here's another nice piece from the paper i think can continue this discussion about probability and maybe also how active inference reframes it so the physics of flow embedded in active inference now it's an ongoing discussion exactly the formalisms that get us to a physics of information flow on sense cognition and action so other live streams other days other years but let's just roll with this physics of flow it's accounting for sensory motor behavior to show that motion in a biologically realistic state space irreducibly includes two orthogonal kinds of motion the irrotational and the solenoidal component so that's what we're looking at here just this is a paragraph that's representing this piece instead of just value only we have value and flow and again this is key to reiterate because it's something that isn't included in reward scalar representations so the former that the first one the curl free is what allows the flow to climb a gradient towards more valuable or probable states now that's one of the head scratchers in active inference that preferred states are likely states under an optimistic deep generative model and that is the alignment that allows us to reduce uncertainty and by way of reducing uncertainty achieve our target outcomes and so when we align the deep generative model in an optimistic way such that we believe that likely states are preferred ones that it's likely that tomorrow i'll have a regular body temperature it's likely that tomorrow i'll be in this preferred body position then we actually get to pursue value as a consequence of reducing there's more

to say on this but dean go for it and then stephen yeah and i think daniel that what that points out is that there's a dependency for people who who who want to introduce statistical propositions to what's the underlying situation that we're observing um there's a dependency on the derivative math and the accounting math you you can't you can't figure out what probably is without figuring out how and what and having a having a a sense of confidence around what how and what are so i don't think it's it isn't just the introduction of active active inference is also wholly dependent on a strong foundation in terms of really being able to get from priors through that hidden state and how do we do that we have to have derivatives and we have to have accounts so i think there's a real codependency back to the between comment that steven brought up cool thanks steven yeah that that that idea of um things going through cycles i think that i'm not going to go into ergodicity again but you know it often comes up is that's from what i'm understanding is is pretty much not present in optimal motor control theory because it's it's about you go there and then now you have a vector and that vector is a thing and that thing gets optimized but you never have that thing i think like dean was saying there's that kind of in-between and you're kind of holding that in between but but that only makes sense if you keep revisiting the in between like if the you need to have ergodicity of whatever i mean that's an open question again for another day but the there's some way that you're revisiting this understanding because otherwise you've got no way to make um any sort of information gain or information extraction unless you have a revisiting to compare interesting um and it's one of the key pieces again that we reduce our uncertainty and achieve preferred states through that process i'm just imagining you know something from organic chemistry it's like you want to study up cram for your final at ochem you want to pass so are you going to take a reinforcement learning perspective or an active inference perspective well the reinforcement learning perspective is like saying you'll get the big reward when you pass so what's the most rewarding problem that we should be solving or what's the most rewarding chapter to review but you don't know because if you knew then you'd pass ochem so you're kind of right back at square zero with value because you framed learning as a value question in contrast the active inference learner would be like oh sure my preference is to pass how am i going to reduce my uncertainty about the critical path of passing well let's i haven't even opened the textbook let's go to the first table of contents or something like that like reduction of uncertainty gives a first step and then it even allows for okay maybe that wasn't even the right first step or maybe that was a good two-step path and now i'm ready to go a different way but i'm trying to reduce my about how to get to what i prefer which is passing rather than i

have a priori valued this future state and now i'm going to back propagate value at every single step of the line and then i'm going to stress myself out if this specific chapter isn't maximally rewarding because everything again has to get reduced to value in this value is everything framework but in the uncertainty reduction framework we see sort of this mist of uncertainty and then regularities and especially dependable ones emerge out of that and so we're planning amidst uncertainty with preferences in mind trying to figure out what to do that's planning as inference amidst which is just a world apart from planning to maximize value through time given you know what model again of how value is assigned it's just a recursive question and value learning doesn't give us that higher level answer for how we should actually go about planning to obtain values let alone critiquing value as a pervasive framework stephen well actually that's a very interesting talk about organic chemistry experiences at university that i had an absolute nightmare on my organic chemistry i might i've given my textbook to someone i've got behind on the organic side and um it was this like two inch thick textbook on organic chemistry and every time you go into it's got so many words it's long names and all of this and what's going on it's really interesting because we had this class with the new professor and he got there and he got a few people to go up to the board and it became clear that absolutely no one knew their organic chemistry and no one had seemed to pick up on this so we were now in the second year no one knew they were doing i think it was even the third year and he said and he said he recommended a little thin exercise book with exercises not on all the different chemical reactions and all the structures that just the key reactions and just to go through these exercises in this thin book and basically that got me through the exam and got me on track not and i you know so going for the big book and the the memorizing and the optimal control it was just work out the five types of reaction and do the exercises and i think that speaks to what you were talking about there you know which ways the electrons move and and that suddenly really really changed everything there's also some fun drilling happening of course all around me welcome back dave um so this section right here it's it's a nice challenge it increases the difficulty of speaking but it's fun this um is a challenge for other frameworks this idea that the two flows together provide a richer framework for expressing skillfulness as a process whose characteristic go beyond a simple gradient of value accuracy so we saw that reward is enough paper recently and some participants in our lab were kind of curious about writing a response because that's the whole argument is they said hey reward is enough the kernel can be reward and that's how you get all of these intelligent behaviors provided that you have these second level um tweaks on the model or these bolted on modules

and then the contention of the authors here is that value functions and indeed any motor scheme based upon functions that return scalars are not up to the task of modeling the variety of skillful acts describing human behavior any kind of behavior just human because by construction they cannot account for the solenoidal aspect of flow so that that's pretty interesting that when we have to bottleneck into a single number we lose a lot of the expressivity that explore and exploit behavior to almost coexist because we're collapsing onto a number line so there's extremely limited availability for you know what do you want for your main dish and your side dish and you just say seven you can't express two things with just one number but if you could say you know seven and three you could express something that's more in a two-dimensional space blue so i think you bring up an interesting point about any behavior or just human behavior um and i think you know there's like dolphins that use tools right to crack open like fish and or to crack open not the fish but the um the sea urchins right so there's dolphins that use tools for eating and then there's dolphins that don't and and so is there and like this is like a hereditary like it's it's an example of cultural evolution in dolphins actually so it's you know it's hereditary but only because it's passed on through learning and social learning right so you know you bring up and and then we saw the motor plant right like the motorized plant like does the plant learn where the like how like skillful performance i mean this is a skill for performance on the part of the plant so it's a really interesting question whether you know i mean we think about active inference as applying to all non-equilibrium steady state systems but does this maybe apply only in the human sense of movement movement or what about like tribes of humans like what about like in a nomadic kind of population would this apply there also cool here's another nice piece in this section that i think will be useful for us to learn about so technically it's easier to read when it's already been written when you're getting the drilling experience that's when it's the hardest to skillfully improvise but reading is a little easier technically active inference extends popular predictive coding models used in neuroscience where perception critical only perception is cast in terms of prediction error minimization so that's the classic round ballard 99 paper that's sort of a mega citation and there's a huge amount of subsequent work and predictive processing or predictive coding is just looking at the relationship from sense to updating of a generative model of sense so it's a signal processing framework and this is something i know that dave is a big fan of both he knows drills and hammers i know that dave is a fan of talking about the signal processing and the audio processing routes of active inference so this is just on the inbound just from sensory information to estimates of that generative model of that

generative process of the sensory data the active inference framework extends this account of predictive processing to model motor control and explains action selection by appealing to the minimization of divergence between predicted sensory data and actual sensory data so here we have the exact same divergence minimization it will sound better on the live stream my jitsi colleagues don't worry um it's the same divergence that's being minimized which is the difference between observed and expected sensory data so in predictive processing it's about sense and perceiving sense and so you want to minimize the divergence between what sensory data you expect and what you're actually getting but active inference adds inference uh planning as inference into the loop and explains action selection also in terms of the exact same divergence goal of predictive processing which is again to have your observed sensory outcomes come into line with your expected data yeah i i one of the things i used to share with people is so today i'm i'm wearing my uh murder of of crows t-shirt because i used to study a lot of corvid behavior and even though i there's there's still some question as to what verbal cues there what instructions corvids are able to share because they are social animals and they're they are problem solvers but you don't have one corbin standing to the side as another corvid solves a problem giving them instructions so i think it's kind of interesting when we're when we're looking at this behavior question whether we're the only species that are actively inferring because i think that there are other species out there that you can make a case that they are doing that probably form of questioning and answering as well um so and and yeah it can be tools it can be based on on on social constructs um it can be based on caching so deep memory and again there's a dependency i think for behavior that uses active inference to also some grounding in how and what so i don't think that active inference in and of itself will allow us to to minimize anything unless we have the derivative and the accounts so i think it's i think it's it's a virtuous triad here but i think what this paper is trying to do is get away from the idea all problems can be solved with only two kinds math cool thank you stephen yeah further to that point around you know what humans do or animals do in general is you could say optimal control approaches maybe it's only really humans that can apply that extra level on top as this kind of constraining kind of deductive idea and every every other organism is really working with this kind of flow i mean if you see a squirrel trying to run up in a tree and between trees and chase another squirrel around the trees it's quite phenomenal just how agile they are and the speed they can work at and um so there's some sort of skillful performance that they've got so there's this interesting thing whereby you've got this we see optimal control as more straightforward but it basically it's almost really optimal control just

isn't implementable in practical terms it's almost it's a it's an what you could say it's a non-ergodic deductive extra that humans bring to the table to explain the mechanical world that they're bringing to bear in some way so it's kind of interesting it just happens to be that was the first explanation we had for this type of stuff nice so let's think about what active inference is and then see who might be doing it or what might be doing it so again the core idea is that rather selecting commands the organism in active inference is inferring what it must should be doing under this same imperative of predictive coding which is to minimize the relationship minimize the divergence between observed and expected outcomes through time crucially this brings perception and action together in the same functional profile we have them on a current sort of trading uh market almost like we can reduce our uncertainty through updating our model and we reduce our uncertainty through action so updating an action or action and are put on common footing and it also explains some so this is the sort of computationalist or um algorithmic level is we're making a sensory motor model that doesn't just have arrows connecting pieces ad hoc and say look they're connected because i drew a picture we're bringing perception and action in a common modeling framework and also on the neurological it explains some of the similarities between the functions of sensory and motor cortices so that speaks to dean's points earlier and to a lot of these brain regions that might have been discovered or initially characterized as having a more cognitive or a more motor role but then it's shown that they play other roles and so you know cue the endless clickbait oh this brain region associated with this also does that should that be surprising should that be salient or should we have another view where that is to be expected now while this move from a problem of so optimal motor control that's the action side and then there's so pure action side tends to not concern itself too much with uncertainty of measurement or deep generative models and then you have the pure sensory processing side that's kind of this move from a problem of control to one an inference in terms of active inference does not make the problem mathematically easier in and of itself but it offers a different model of skilled action which respects the neurophysiological evidence so in other words we're taking the sort of problem of going from sense data to a generative model the predictive processing model and then the forward model from internal states to action selection and it's easier to of course subset to just one of those directions and say that one has a tractable model but active inference is combining both directions or integrating across these two ways of like looking at sense in action and saying let's have one model of sense in action with a common functional profile as they describe it here now which systems have it slash do it that's the realism question and the

instrumentalism question is which systems can we model as if they're doing this so i think that the instrumentalism question which as always it's about us so it's easier to answer is the kinds of systems that have sensory motor input output dynamics so everything from a self-driving car to different kinds of animals we could model as if they were getting input generative model output so the things that people model with ooda or the that classical sandwich of cognition those are the kinds of systems that are apt to be modeled with active inference the kinds of things that people were modeling with optimal control going from the model to action and the kinds of systems that people were modeling with predictive processing sensory inputs to generative model both of those and ones that especially are doing both those are some of the amenable uh systems for active imprints stephen you know i was wondering this this idea of functional profile is i haven't heard that one before it might be another one to think on the ontology that's been talked about how have you seen that being used anywhere else um in terms of active inference functional profile good question it doesn't seem like it's used anywhere else in this paper and i don't off hand recall what other papers it might have been used in but that's a great point something to ask about um i think that the notion of just function is a key one to follow and uh what what how how do we follow a function yeah go ahead stephen yeah i suppose because this is talking actions skillful performance function starts to it's thinking about functional as in how to achieve an action of skillful performance so maybe that's how it comes in here whereas a lot of some of the other papers there's no moving parts so to speak it's all information based in in terms of um but i think it's it's interesting in terms of um what that does or what it's trying to do it's another way of trying to speak to something which is a bit bigger you know that's one thing i've noticed inference normally things are trying to define what everything is this one it gives these it gives these pullbacks they're like okay we've got a functional profile now that allows us to um do xyz but like i say it does provide that challenge of trying to fit that into these other pullbacks that we seem to be keep finding appearing like you know the effective charge that came up in one and so yeah i think that's a kind of an interesting question about how that fits in with other types of models or ways of translating between spaces or frameworks dean thanks yeah so i wonder so let's just take that to its its next logical place so let's start with a functional profile not necessarily work back but is there some way for us to be able to identify what's in that profile that would allow for active inference and should we is that something that that is that dangerous or is that legitimate nice question here's a little bit where they're getting at this realism and instrumentalism discussion so even some of the most forthright defenders of the active

framework assemble have backed away from making such strongly realist claims about the causally efficacious character of generative model so here is where we see um hardcore realism so it's nice you know and as we work towards the terms and the ontology how can we classify different senses or uses of terms first in 2013 has advanced the view that an agent does not have a model of its world it is a model now that's an interesting claim um but the authors bring this up by also qualifying Friston's use of model from a different paper and saying crucially if we accept the generative models of the active frameworks not sure if this is the if there's saying you know usually it's framework but are will there be multiple active inference frameworks in line with these proposals we must embrace the idea that generative models are causal in the sense of an agent's inactive attunement with the world rather than part of the causal efficacious machinery used by an agent so that's an interesting claim blue so i i was thinking about like you know that an agent doesn't have a model of the world but it is a model and like i think about that in terms of people that are that get have traumatic brain injury and then have like dramatic amnesia like can't remember how to walk or talk or eat and have to like relearn all of these things right so so i think about like if we don't i mean in that case it seems like the like you have a model and then you suddenly don't have a model but it's like if you are a model are you suddenly not a model or is your is that like a temporal disruption i don't know like so that that to me i mean and this is an instrumentalist realist debate i realize but where do you you know if you are a model is it like a temporal disruption and suddenly you have to start over or what does that that look like in in those circumstances nice question stephen yeah i mean in a way i suppose when someone has to re-learn that the fact that they are at some level their body is part of being that model that gives them access to either some priors or that they call it the morphological computing capacity the body's capacity to then form those priors because um you know if if however certain parts of the body are like with a stroke that can be kind of challenging because that maybe the actual way the body moves is compromised therefore the ability to use it to recreate is is diminished i mean the question of how much it's diminished then becomes i don't know if anyone could quantify that but it's interesting it's a good point an instrumentalist would say well yeah the brain isn't you know per se doing active inference we're going to model it as such but then when there is an injury to the physical system then we say oh we're still just modeling it with active inference we're just changing our active inference model or has the actual system itself changed so will we just always say that even when the system changes in the physical world we're just going to update our model and just always stay in the instrumentalism land or are there times where changes in

the world make the instrumentalism verge into realism as if it's actually describing something about the world that's going to be a very delicate walk and they they are arguing that again if we think about these generative models as purely formal and epistemic tools so just like a linear regression you know you can do a linear regression on gdp or on some other feature you can do a t-test in a double-blind study it doesn't make the people a t-test it doesn't make the gdp a linear model so it doesn't make something active just to use it in this statistical that's what's meant by the formal and epistemic tools then this framework is we argue compatible with a sensory motor approach to understanding skilled action it's interesting to me because the sensory motor approach to understanding skilled action itself could have a realist or an interp or an instrumentalist approach the instrumentalist sensory motor would say i'm not even saying this system has sense in action i'm just saying we could model it as if it had sense in action whereas the realism of sensory motor would say this system actually does have sensory and active components and also this is another important note the sensory motor view reflects a spectrum of ideas which includes simple open loop sensory motor correlations closed loop systems regularities given a goal and optimal sets of regularities according to a certain performance metrics these can be understood using the tools of dynamical models of capturing the brain body environment interactions in terms of dynamical systems as opposed to assuming the agent is involved in symbolic computation so it's almost like we see a lot of separation of uh the isolation of agency or the delineation of agents with agent in the niche and then the niche itself so we see like agent and external internal states and external states and now when we think about brain and body that's one sort of inseparable dyad and then it is also inseparable from the niche so how are we going to take all these dynamical systems which have different physical and causal connections amongst them brain body and niche and work towards a framework that's actually going to say what we want to say and be useful for what we want to do with these kinds of models like reducing uncertainty about how organisms will act or designing environments that work different sorts of agents how are we going to do that effectively and drawing from what we know about dynamical systems more broadly rather than bring on assumptions such as that the agent is performing symbolic computation which comes along with the instructionism and the representationism and all these other pieces stephen you know this is a useful uh sentence here to show that dynamic that's happening between the kind of participatory sense making world um and the sort of inactivists and the inactive inference world um because there's there's these useful distinctions that have been coming up in an activism and this participatory sense making yet they struggle to give the

ways to actually calculate how that comes about outside of these kind of metrics that are looking from the outside of the system so um this this is kind of quite a dance that's going on but it seems to be yielding some good um progress for active influence because they're laying some of the ground for active inference to then see what could be plausibly uh modeled so that's quite useful yeah another point to that and also what you said they're suggesting that active inference is going to extend costly linear accounts of motor control which tend to focus on the contingency between new actions and their consequences so the loop there which is hardly a loop it's actually a line goes from action selection to a new state like i want you know i want to grasp the coffee cup how am i going to grasp i grasped that the domain of many previous models which is like from some sort of delta that needs to be accomplished to achieving that action and that's where the model ends active inference in fact proposes a complementary view where predictions of expected proprioceptive states are not just seen as passive reactions to new motor signals like in the coffee cup grab example but as also triggering adjustable dynamic reflex arcs to generate new actions so here instead of starting from action to a new proprioceptive state like i want to be in a different state so um i want to select an action okay now my elbow is in the correct predicted proprioceptive state so we're good to go here we're starting with the updating of our proprioceptive state moving to action selection inferring our state so action and like the two-stroke engine are now working together in a way where we can pull out a little local stretch that has the same functional features as this linear causative model above because there's going to be some stretch where that action was performed and where the functional characteristics were carried but we're embedding that in a deep temporal model and the temporal depth of this model confers a more active anticipatory role to proprioception now seen in a causally circular model in line with the inactive and embodied so that's that sort of rummaging around that gives an anticipatory and active role for not just a state estimate role for so that it can get relayed to the central headquarters and sent back as an instruction it's not like a central intelligence model of action selection it's like an anticipatory deep temporal model that allows sense to be doing something slightly different um just a question when it says and so it says new action so new action in active influence is the new well guess i suppose of what the action state is and then the new proprioceptive state this is interesting thing with is is that almost a guess as well like based on a mix of um sort of intrceptive data and other data or data but other ways so is it like a different types of guessing or is one of them action ones still being seen as perception in that loop good question i'm sure there's a lot of ways to go about it but one thought is even perception is inference

that's predictive coding so our visual perception is clearly a generated inference there's no blind spot there's resolution and color in the periphery where we know the anatomy of the eye doesn't directly support that and so on there's smoothness even though our eyes are constantly saccading so whether it's a proprioceptor or whether it's visual input that is also inferred as part of the deep generative and so when there is ambiguous visual input or ambiguous proprioceptive input for that inactive inference is an opportunity for uncertainty reduction rather than getting out over our skis within saying whether that's a valuable a less valuable state we actually don't need to judge the value in a scalar framework of ambiguous versus unambiguous states because we can just pursue uncertainty reduction and then also cast planning as inference and perception as inference does that make sense stephen and then blue yeah that's that's really useful and i think it's like what you're saying there that also brings in the the idea of temple scale so there's with vision and with this there's what's going on below at greater speeds and flux than we can know like the circades and the way those circades then get put together to be this kind of joined up smooth perception of vision and this this is this new action new proprioceptive states that's almost kind of at the speed of everyday awareness you know and once things become longer and happen over the period of like 10 minutes intervals you can no longer track that in your working memory or your working phenomenological consciousness so this is kind of like the goldilocks zone in a way where we're able to when it's talking about new action new proprioceptive states it's kind of sitting at a kind of um an averaged out perceptual awareness in a way that we can engage interesting point blue yeah daniel you already like kind of said what i was going to say right after stephen but um proprioception really is is the expectation like where do i expect my body to be like instead of asking like where is my leg it's like where is my leg supposed to be or where do i think my leg actually is and you're not always right right like in the phantom limb um case and even like you know if you're stuck with a pin right like so and this is not proprioception this is just generally like where your nerves are so some in some places they're very close and you can infer with great accuracy where exactly the pin was poked into you but in some other places in your body the nerves are more far apart and so you can't tell even if the nerve even if two pins you put a pin you poke a pin here you touch the same place it's like the pin is here but you're not always right so so there's there's this inference that that happens um in as well nice it's like there's um one mode where you have a very high confidence prior of where your arm is located you know it's right by my side let's just say and when there's a high confidence prior data that don't support that prior are only going to move it slightly if someone's very

convinced that something is the case and they say it's 100% the case that this is this way well then you know there's no point in providing alternative evidence because they're going to see it in terms of their prior so in that case of the phantom limb it's like there's a high precision prior that the limb is in a certain location in fact it's such a strong prior that sensory data like proprioceptive data that your arm is in a different place or visual data for example somebody no longer has that limb still doesn't manage to update the experience or the generative model of the limb being there so that's one case that we can frame in an active inference way and another case would be like where's my leg i have low confidence and now instead of framing this just as an estimate that there's an uncertainty on and then we're going to have to tag a value on we go from that estimate with uncertainty to planning as inference what motor control or what motor behavior is going to reduce uncertainty on the leg and sometimes that's like looking at it other times it but that's a motor behavior with the eyes other times it might be doing a motion that then like kind of uh gives us some sensor data so it's kind of i don't know like checking your phone and seeing like what notifications are there when you're uncertain maybe sometimes like a shake or something like that it's like just reactivating a bunch of sensors just to pass say oh yep we're here just checking in reducing your uncertainty and so the state estimate even when there is uncertainty about it we can reduce that uncertainty through rather than just being trapped in a framework of like okay i'm not sure where it is and now i'm not sure whether that's rewarding or and now i'm not sure which path to take that's going to be most rewarding we don't go down the reward rabbit hole because there's a deep generative model for evolved slash created or whatever organisms those priors are fit so it's not just like we're dealing with the space of all possible generative models these have been evolutionarily selected to be functional in our physical constraints but underwater or in a different gravity space it'd be a different set of priors that would need to be uh learned or utilized in our last piece here what what do you think were some of the big pieces that we leave this 23 strange attractor with as we move forward like what kinds distinctions or insights here do we want to see how they apply in different settings dean and then anyone else okay so this is gonna be a strange twist but that's that's okay so i came away from the and i thought about some of the things casper did bring up casper hess did bring up in terms of so so what happens when when we talk to instructionalists like instructors are all around us there's people that spend a great deal of their life preparing to become an instructor and we come along and say um yeah that's great we know what you do um but we're gonna i don't know maybe bring you down a peg because we're going to introduce this stuff that you weren't asked to sort of train yourself up

on and i'm not sure if that's a threat or i guess it depends on the personality who the idea is introduced to but i'll bring i'll bring up if you know the movies independence day and thelma and louise um i'm going to bring those into kind of into a juxtaposition here i think for the instrumentalists um they don't necessarily see that there can be for the person who's living out the adversarial greek myth us against the aliens or us against the the people who are chasing us down and trying to get our our convertible back um there isn't there is no set of instructions per se that those people want to follow they're living the interactionist state and you could start out with kick the tires and light the fires all uh harry connick jr and still end up in a rapid deceleration moment a la susan sarandon and and gina davis and i think for people who are going to look at this problem in terms of how do we introduce interactionism in a way that doesn't threaten the top-downedness of how most most situations where we want people to be able to take up this embodied skill skillful performance without putting them in a in a situation where they behave like aliens or behave like police chasing somebody down with a helicopter we're gonna have to sort of i don't know i don't know if it's a slow rolling i'm not sure exactly what it is but those people who are now currently instructors and instructionalists by by profession we're going to have to give them something so that they don't cling desperately to a model that i think the authors of this demonstrated requires a rethink i don't think we throw out instructionalism but i absolutely do think that it's its place on the top of the mountain and now all these little robots climbing up it that that has to be reviewed and revisited and so i mean as a go forward i don't know how we put people in a place where they're comfortable with that kind of i don't want to say it's disruptive but i think it is but i think we have to be able to come at this in a way that doesn't make them sort of circle the wagon solenoidally and try to protect something that obviously is up for a bit of evolution before a revolution nice points there stephen yeah i like that idea yeah this challenge of having this mountain that we've kind of ascended and placed these optimal control theories on and maybe it's it's more like there's a mountain of active inference and there's a there's a little rock a rock structure on the top which is optimal control theory which we're able to sort of utilize on top of that you know but it's uh it's it's a lot more rickety and it's not the the main game in town you know the main game is our intuition our dynamical um balancing of all these different free energy parameters um but we do and particularly humans have this ability to impose this extra um shack of tools on the top that we now think is the main game in town nice so to stevens then to dean's point so as to which one's the uh mountain which one's the mole hill which one's the ant hill here fristen says optimal control can be cast as active inference with three simplifications so as we increasingly

clarify our understanding of these different models will be able to see which models are nested within one another or are alternate framings of one another or are you know more overlapping than different or what are those key distinctions or what are the variables where you can expand it to three and all of a sudden you've gone from model a to b so remains to be seen and by uh him saying that optimal control is a thrice simplified active inference along these three specific domains it does suggest that the bigger picture might be active inference and then you get special case optimal motor control with a value function as a derivative model within an active universe even if these other models were developed first that's often the case that a certain model is introduced and then it's later generalized in several ways so that in the future it's actually seen as a special case model even though it's actually at one point it was the only game in town certainly for a lot of biology models that's the case and then dean i really like this idea of like there's a job title called an instructor and if somebody says i'm an instructor you know organic chemistry or in tango baseball people know what you're talking about so how could interactionism be so broadly participated in that people say yeah i'm an interactor i'm a baseball interactor or i'm an ochem interactor maybe the job would be called that or not so how do we like you were saying not kind of threaten or um disrupt their modes but give a yes and for what they're working on and maybe part of it is that the interactor it's like we're framing their interactions with their student let's just say as a cultural scaffolding of an interaction that doesn't make it a symmetric interaction it's a culturally scaffolded moment of bayesian model updating and so it's like yes you do give instructions that's the verbal term for what you say when you say pull or breathe or okay slow down you can still give instructions we're not saying that that is not part of your toolkit but let's more broadly look at your interactions with this dynamical system as doing what you want to do and maybe it'd be enough just to push them to that edge of the hill and just say interactions are a broader way than instructions would you want to reduce everything you do to just barking instructions or sometimes you lead by example other times you show wordlessly you don't need to frame your interactions as instructions recognize that you're just dynamically interacting systems and this a lot of people are working on a framework that suggests that this is a broader way of thinking about training and might be able to help you on one hand with the high performance elements and on the other hand with accessibility and in making your work more communicable not sure about that of course but just one way to go steven yeah with the idea of roles and how people uh look at performance in roles this really comes back to coaching again i mentioned that earlier is you know we've coaching came out of the need to supervise

therapists and how they could be guided in their path as they were supposedly helping the people that were needy they needed guidance and that same idea now of coaching rather than teaching or instructing and um while they have their role as giving the formal documents that you might need providing the bigger narrative and giving certain tools you know you you within that you there's this idea of okay what types of developmental or interactions can be nurtured and cultivated and i think this actually really can tie into the kind of pragmatist social constructivist approaches that coaching's trying to work with as in but it gives it a bit more um does actually give it a bit more meat without becoming too complex which i think previously a lot of the active influences may be a bit too abstract but this is starting to show a bridge there of why coaching at different levels can and positive psychology more broadly can have a role okay great thank you stephen dean yeah stephen i appreciate exactly what you're saying i i think to get to a place where is no longer threatening people have to want or appreciate that you can be a subject matter expert and a prediction matter expert and i'm not sure i think there are maybe are some coaches who are both and i but i sort of transcended that and got into the wayfaring thing because wayfaring imposes a prediction aspect to this now i can't predict for you whether that running drum shot that you make i can't be in your ear i cannot coach you to take that shot in that moment with that much time left on the clock in fact if i do that i'm probably adding impediments and i'm being an optimal control list micromanaging you and you whether you make it or not it's not about me it's about whether the ball and the and the and the basket arrived at some sort of outcome that we both want i think that i think that that's part of how we get people to a place where their dependency on their subject subject matter expertise doesn't overwhelm the fact that for these performances there is a predictive element and so we have to sort of give that equal billing i don't think we have to over bill one or the it's back to the between and that's a prediction matter expert and again i don't know if it's a role being away fairer because you're in the boat with that other person right it doesn't matter whether you're the one saying okay well i'm i'm getting a sense of the direction of the current right now and the other person's looking up at the stars you're working and so you everybody brings their expertise and i think that that only that only materializes if we can if we can include something with the subject matter expertise culturally yes absolutely there's huge cultural piece to it but i think i think you can be a coach and not necessarily be a wayfarer but i think if you're a wayfarer you have to bring a predictive element into it otherwise you're not being a wayfarer interesting about having an integrated way to talk about goal and process and interaction with a niche interaction with your co wayfarers just like your example with a

movie you know there's is the map app going to tell you which directions to take that's kind of on the instructionist end versus are you on the run so anywhere you go you're you're just not following instructions you're just negotiating sequential interactions and reducing uncertainty in a way that isn't just looking down at the application and seeing which way to turn when nice i'm glad you picked up on the uh the map going out the back end over the trunk of the car exactly oh you're reading it oh no right um cool well steven yes yeah just one question way fairer is that like a particular sort of approach is like an orientation based or well yeah yeah you're on your path and how are you faring how are you feeling okay right it's not just way finding it's not just the discovery piece but it's so how does that make you feel being in that uncertainty space because again if we're going to close loop discovery will eventually loop back around to recapitulation and then all right we're open systems that are constantly closing these loops so how does that make you feel right okay to connect that also to john boyk and second order science as well as a bucky fuller in trim tab when we're the rudder we need to act in an fashion so that's kind of like first order cybernetics the rudder has to turn before you hit the iceberg you know you gotta find a path so pathfinding and that whole notion of anticipatory comes to the rudder and the trim tab is the little rudder on the rudder so it's anticipating even the anticipatory nature of planning because you have to turn that trim tab even before you turn the rudder and so here we have instead of a sort of mechanical model with recursive levels of rudders we actually have hierarchically nested priors and so it's a little bit like higher and higher level priors that in one sense when you look at one and you try to tell a story by backfilling what happens they do act in increasingly anticipatory ways but then when it's seen as just part of the wayfinding of the ship they're all in a cyclic feedback with each other and it's not like one is really upstream of another but there are certainly pieces doing different things cool stuff any um final thoughts or questions i don't see any other questions from the live chat but another um interesting and surprising as usual does anyone have any final comments well great we'll um look forward to some awesome upcoming streams let's just look at those really fast next week on the 21st we're going to have our symposium with carl fristen so if you've been helping to co-organize this event and develop the questions with us you'll be with us live otherwise we're going to be recording this not live streaming it and then providing the recordings on our youtube channel and then for the second two weeks of june we're going to be doing a more technical or statistical empirical evaluation of active inference in multi-armed bandits then in early july we're going to have a fields and living paper scale free biology integrating evolutionary and developmental thinking then uh

don't have any papers for the end of july or beginning of august but we'll figure out what we want to do there and then we'll be turning to some industrial engineering and some other areas so as always pretty fun journey thanks uh dean blue steven dave for participating live and everyone watching live on replay we'll see you on a future stream